01 · PROBLEM AND OBJECTIVE

Foundation UI

An accessible, token-driven React component library, proven in a working app and published to npm.

Repeated work

Every app rebuilds its own buttons, forms and dialogs, each slightly different.

Uneven accessibility

Labels, focus and error messages depend on who built the screen.

Hard-coded colours

A rebrand or a dark mode means editing every app by hand.

Objective: one package, one source of design tokens, proven in a real employee dashboard.

Scope and architecture

44

components in six groups

Basics	12
Forms	10
Overlays	7
Layout	5
Navigation	5
Data and dates	5

packages/tokens

Imports the Figma export, validates it and builds light and dark CSS variables. Private.

↓

packages/ui

44 components, one ES module each, plus compiled CSS and types. Published to npm as @rachitp72/foundation-ui.

↓

apps/demo

Employee dashboard that imports the package by name, exactly like any other app.

Design tokens and themes

Figma
Tokens Studio and Variables, owned by design

→

Export
One tokens.json file in the repository

→

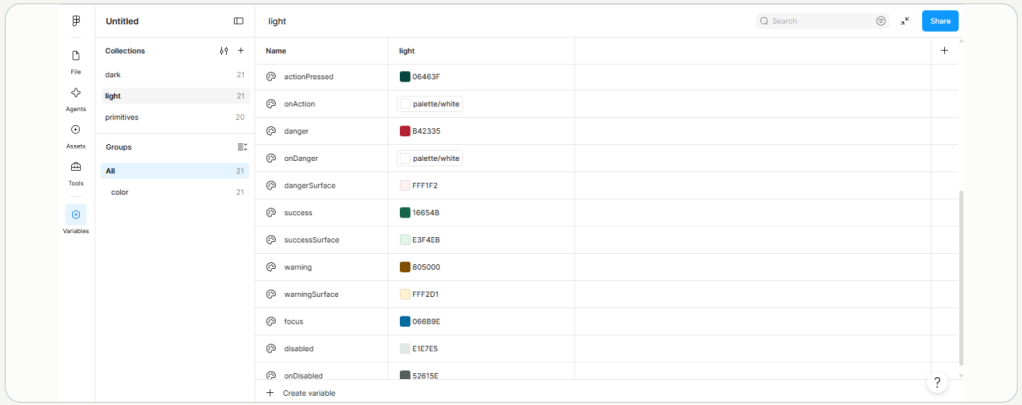
Import
Validates it and lists every change

→

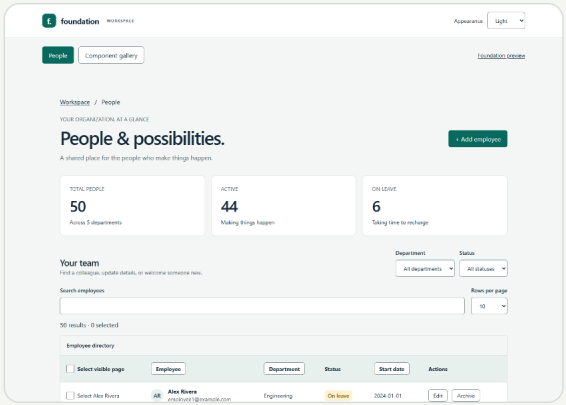
Build
Style Dictionary generates the CSS

→

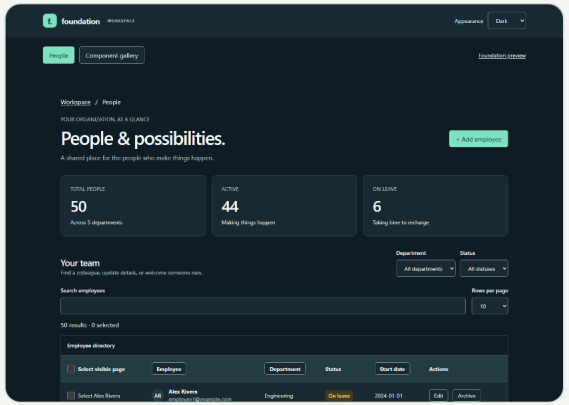
Themes
light.css and dark.css as --fui-* variables



Token variables in Figma



Light theme



Dark theme, same tokens

04 · HOW IT'S BUILT

Implementation approach

Native controls first

Real <button>, <input>, <select> and <progress> elements, so keyboard use, forms and autofill work by default.

Proven primitives

Radix handles focus trapping and keyboard rules for dialogs, menus, popovers, tooltips, tabs, accordion and toast. react-day-picker drives the calendar.

Own typed public APIs

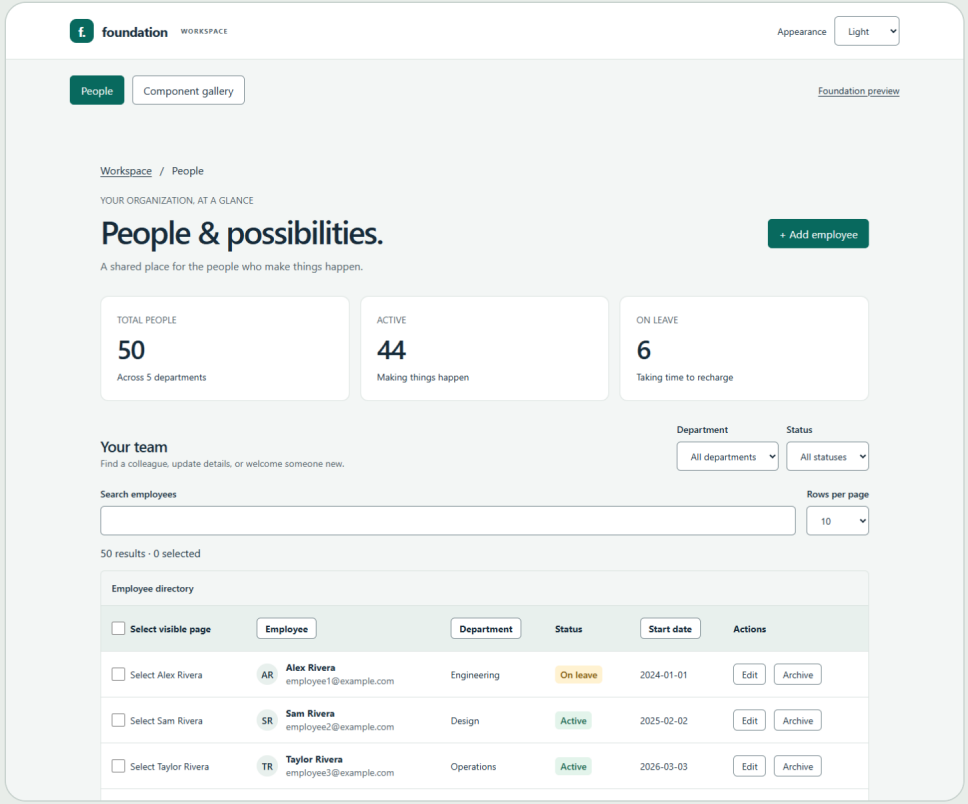
Each component wraps those libraries in its own props and token styles. Inputs work controlled or uncontrolled; dates are YYYY-MM-DD strings that never shift across time zones.

Reuse by composition

```
DatePicker = Input + Button + Modal + Calendar
DataGrid = Table + SearchInput + Checkbox + Select
+ Pagination
```

05 · PROOF

Working demonstration



1 Search and sort

Type “Sam” in the directory: rows filter and the result count is announced.

2 Form validation

Save an empty Add employee form: errors appear and focus moves to the first one.

3 Modal and date selection

Pick a start date in the calendar dialog with the keyboard; focus returns to the field.

4 Feedback

Save: a toast confirms it. Archive: a confirmation dialog, then Undo.

06 · PROOF

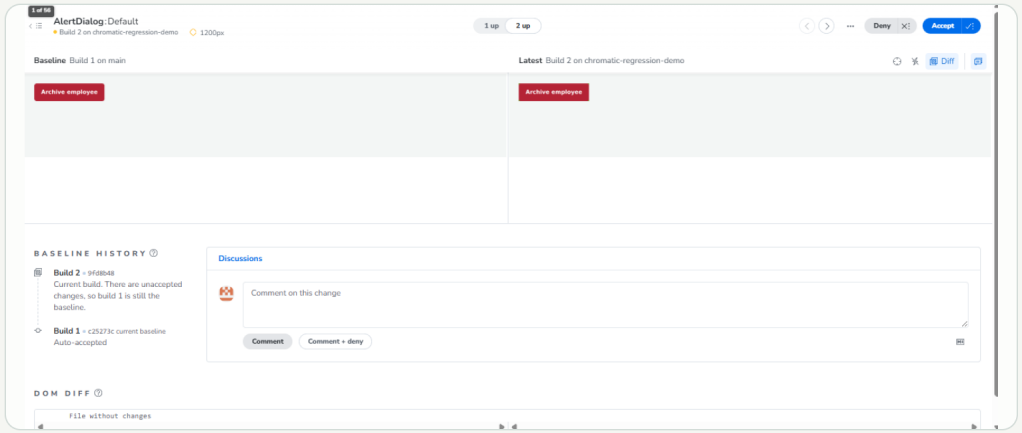
Accessibility and tests

18/18
NVDA screen-reader checks passed on the dashboard

204/204
Storybook stories with 0 axe violations in Chrome

97.4%
statement coverage from 72 Jest tests

57
visual changes caught by Chromatic in a planted regression



What each check covers

- Manual: NVDA and Chrome on 18 dashboard tasks (15 components), plus 400% zoom.
- Automated: axe on all 204 stories; 13 keyboard tests on 12 interactive components.
- Visual: every change is reviewed by a person, never accepted automatically.
- Not covered yet: VoiceOver, JAWS, Firefox and Safari.

07 · RESULTS

Challenges and results

Bundle size

Problem

A single-file build put the whole library into a Button-only app: 95 calendar modules, 389 KB.

Fix

One ES module per component; React, Radix and the calendar library stay external.

Result

0 calendar modules, 224 KB. Button adds 0.3 KB gzipped, DataGrid 3.0 KB (budgets 1 and 5 KB).

A menu with the wrong name

Problem

DropdownMenu's label did nothing: screen readers heard "Actions", not "Employee actions".

Fix

Caught by an interaction test. Label fixed; the unit test that missed it now fails without the fix.

Result

The right name is announced; 13/13 interaction tests and 204/204 axe scans pass.

Also measured: Lighthouse 100 / 100 / 100 / 100 on desktop · DataGrid with 10,000 rows updates in under 6 ms, search included

Limitations and next steps

Honest gaps

- No backend: demo data is in memory and resets on refresh.
- Screen-reader review covered dashboard flows only, in NVDA and Chrome.
- The demo runs locally; it isn't hosted yet.
- One date format; no localisation.

Next, in priority order

1. Run Chromatic on every change; today it is started by hand.
2. Host the demo so reviewers can open it.
3. Screen-reader review of the other 29 components, in VoiceOver and JAWS too.
4. Localised date formats in DatePicker.

GitHub: github.com/rAcHiT1234766/F3-02 · Storybook:main--6adb1ea72d119416b1620b45.chromatic.com · npm: npmjs.com/package/@rachitp72/foundation-ui

BACKUP A · FOR QUESTIONS

How quality is checked

Check	How	Result
Lint and types	ESLint and TypeScript on all code	Passed
Design tokens	Validation and Figma round-trip tests; contrast pairs	10 tests and 28 pairs pass
Behaviour	Jest and Testing Library, 11 suites	72 tests; 97.4% statements
Keyboard flows	Storybook interaction tests in Chrome	13 of 13 pass
Accessibility	axe on every Storybook story in Chrome	204 of 204, 0 violations
Real workflows	Browser tests: add, edit, archive, undo, 320px screen	Passed, both themes
Visual changes	Chromatic snapshots, reviewed by a person	57 planted changes caught
Package	Installed from npm into a separate app	Builds and works
Screen reader	NVDA 2026.3 and Chrome, 18 dashboard tasks	18 of 18 pass

BACKUP B · FOR QUESTIONS

Performance

Lighthouse, employee dashboard

Profile	Performance	Accessibility	Practices	SEO
Desktop	100	100	100	100
Mobile	99	100	100	100

Size added to an app

Import	Gzipped	Budget
Button	0.3 KB	1 KB
DataGrid	3.0 KB	5 KB

DataGrid with 10,000 rows: update time in ms, before and after memoising (median of 7 runs)

Version	Mount	Search	Refine search	Next page	Select
0.1.0	5.9	15.7	16.5	6.5	5.6
0.2.0	4.6	4.9	3.0	3.6	1.8

BACKUP C · FOR QUESTIONS

Using the library in an app

```
import { Button, DatePicker, FormField, Input }
  from '@rachitp72/foundation-ui';
import '@rachitp72/foundation-ui/styles.css';

export function NewStarter() {
  return (
    <form data-ui-theme="light">
      <FormField id="name" label="Full name" required>
        {(props) ⇒ <Input {...props} name="name" />}
      </FormField>
      <DatePicker id="start" label="Start date" />
      <Button type="submit">Save</Button>
    </form>
  );
}
```

Install

```
npm install @rachitp72/foundation-ui
```

What the app gets

- One compiled stylesheet: no Tailwind or Style Dictionary setup.
- Only the components it imports are bundled.
- TypeScript types for every prop.
- Light or dark theme on any container.
- Needs only React and ReactDOM 19.

BACKUP D · FOR QUESTIONS

All 44 components

Group	Components
Basics (12)	Button · IconButton · Link · Text · Heading · Badge · Avatar · Separator · Spinner · Skeleton · Progress · VisuallyHidden
Forms (10)	Input · Textarea · Checkbox · RadioGroup · Switch · Select · FormField · Fieldset · SearchInput · Slider
Overlays (7)	Modal · AlertDialog · Drawer · Tooltip · Popover · Toast · Alert
Layout (5)	Box · Stack · Grid · Container · Card
Navigation (5)	Tabs · Accordion · Breadcrumbs · Pagination · DropdownMenu
Data, dates (5)	Table · DataGrid · Calendar · DatePicker · Combobox

